

Roll No.....

ODD END SEMESTER EXAMINATION DECEMBER - 2025

Name of the Course: B. Tech (EE)
Name of the Paper: Control Systems

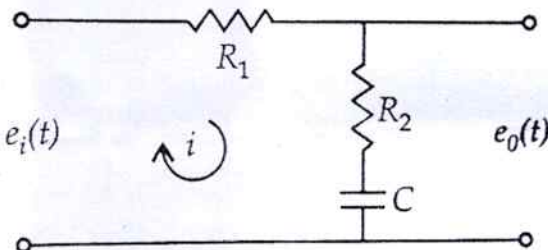
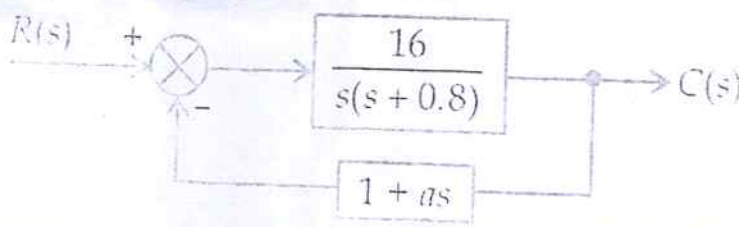
Semester: 5th
Paper Code: TEE 502

Time: 3 Hours

Maximum Marks: 100

Note:

- (i) All questions are compulsory
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty
- (iv) Each question carries 10 marks

Q1	(20 marks)	
(a)	Explain any three types of Translational Mechanical System with suitable diagrams and equations.	
(b)	Determine the transfer function of the Figure. 	CO1
(c)	Explain the Step & Parabolic functions with suitable diagrams and Laplace transform of that input function.	
Q2	(20 marks)	
(a)	Explain the following terms: (a) Settling time (t_s) (b) Maximum overshoot (M_p) (c) Delay Time (t_d)	
(b)	Consider the system as shown in the Figure. Determine the value of a such that the damping ratio is 0.5. Also obtain the values of rise time and maximum overshoot M_p in its step response. 	CO2
(c)	Derive the expressions and draw the response of the First order system with unit step function input.	
Q3	(20 marks)	
(a)	Derive the expressions and draw the response of the Second order system with unit step function input and underdamped case ($0 < \xi < 1$).	CO3,

(b)	Define the following terms (if necessary draw diagrams): (a) Close-loop control systems (b) Static acceleration error constant (K_a) (c) Type 1 System with example.	
(c)	The open loop transfer function of unity feedback control system is given by $G(s) = \frac{50}{(1 + 0.1s)(s + 10)}$ Determine the static error coefficients K_p , K_v , and K_a . Also determine the type of the system.	
Q4		(20 marks)
(a)	Explain the Bode plot concept in the control systems? Derive & sketch the Bode plot for the transfer function: $G(s) = \frac{1}{s^N}$ (Where, $N=1, 2, 3, \dots$)	CO4
(b)	Derive & sketch the polar plots for the Type 'Zero' control system.	
(c)	What is the frequency domain analysis? Explain with suitable equations. List the advantages and limitations of frequency domain analysis.	
Q5		(20 marks)
(a)	Explain the proportional control action with suitable figures and equations. Limitations and advantages also explain.	CO5, CO6
(b)	Define the following terms (if necessary, draw diagrams or write equations): I. Phase Lag Network in Compensation Technique II. Advantages and Limitations of State Space Techniques	
(c)	Construct the Routh array and determine the stability of the system represented by the characteristic equation $s^4 + 2s^3 + 3s^2 + 4s + 5 = 0$.	